Explicit weights on $\mathbb{N}$ and $\mathbb{N}^2$ that are not almost monotone

Candidate full answer to item (4) of arXiv:2407.19982

26 August 2026

Status. Candidate full answer by explicit counterexample, likely valid, awaiting specialist review. The source question has an affirmative answer: there are weights on both $\mathbb{N}$ and $\mathbb{N}^2$ which are not almost monotone.

1 Source question and definitions

Dabhi calls a map $\omega : \mathbb{N}^2 \to [1, \infty)$ a weight when

$$\omega(m_1 m_2, n_1 n_2) \leq \omega(m_1, n_1)\omega(m_2, n_2).$$

It is admissible when

$$\lim_{r\to\infty} \omega(m^r, n^r)^{1/r} = 1 \qquad ((m, n) \in \mathbb{N}^2).$$

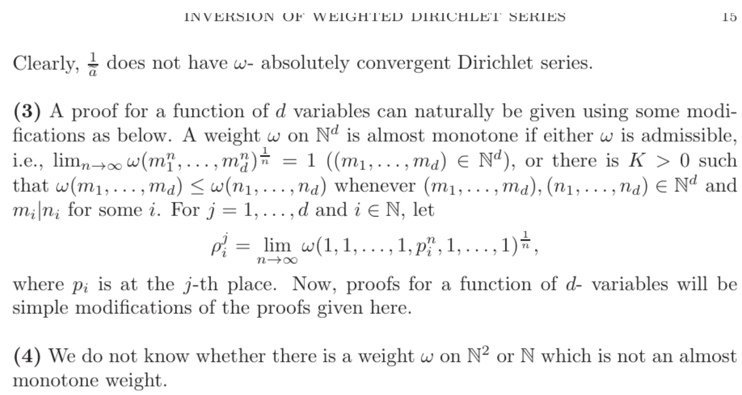
It is almost monotone when it is admissible, or when there is $K > 0$ such that

$$\omega(m_1, n_1) \leq K\omega(m_2, n_2)$$

whenever $m_1 \mid m_2$ or $n_1 \mid n_2$ [1]. For $\mathbb{N}$, the analogous definitions use $\omega(ab) \leq \omega(a)\omega(b)$, admissibility along every sequence of powers, and the divisibility relation.

In concluding remark (4) on page 15, the paper asks:

> "We do not know whether there is a weight ω on $\mathbb{N}^2$ or $\mathbb{N}$ which is not an almost monotone weight."



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Clearly, $\frac{1}{a}$ does not have ω- absolutely convergent Dirichlet series.

(3) A proof for a function of d variables can naturally be given using some modifications as below. A weight ω on $\mathbb{N}^d$ is almost monotone if either ω is admissible, i.e., $\lim_{n\to\infty} \omega(m_1^n, \ldots, m_d^n)^{\frac{1}{n}} = 1$ $((m_1, \ldots, m_d) \in \mathbb{N}^d)$, or there is $K > 0$ such that $\omega(m_1, \ldots, m_d) \leq \omega(n_1, \ldots, n_d)$ whenever $(m_1, \ldots, m_d), (n_1, \ldots, n_d) \in \mathbb{N}^d$ and $m_i | n_i$ for some i. For $j = 1, \ldots, d$ and $i \in \mathbb{N}$, let

$$\rho_i^j = \lim_{n\to\infty} \omega(1, 1, \ldots, 1, p_i^n, 1, \ldots, 1)^{\frac{1}{n}},$$

where p_i is at the j-th place. Now, proofs for a function of d- variables will be simple modifications of the proofs given here.

(4) We do not know whether there is a weight ω on $\mathbb{N}^2$ or $\mathbb{N}$ which is not an almost monotone weight.

Figure 1: The multidimensional definition and item (4) on page 15 of the source paper. The earlier $\mathbb{N}^2$ definition on page 2 includes the uniform factor K used above.

2 Proof intuition

The construction separates two requirements that might at first seem to force each other. A linear term in the exponent of the weight produces nontrivial spectral growth and hence nonadmissibility. A second, sublinear term creates unbounded downward jumps along divisibility.

For the second term, write an integer in base 4 and evaluate the same digit string in base 2. This digit cost is subadditive: when two base-4 expansions are added, every carry replaces four copies of cost 2^i by one copy of cost 2^{i+1}, so the cost decreases. Nevertheless, just before a power of 4 all digits are 3, while at the power of 4 only one digit is nonzero. This creates an arbitrarily large downward jump. Combining the linear term with this digit cost gives exactly the two properties needed.

3 The digit-cost lemma

For $r \in \mathbb{N}_0$, let

$$r = \sum_{i=0}^{J} d_i 4^i, \qquad d_i \in \{0, 1, 2, 3\},$$

be its base-4 expansion, and define

$$c(r) := \sum_{i=0}^{J} d_i 2^i.$$

Lemma 1. *The function $c : \mathbb{N}_0 \to \mathbb{N}_0$ is subadditive. Moreover,*

$$\frac{c(r)}{r} \longrightarrow 0, \qquad c(4^j - 1) = 3(2^j - 1), \qquad c(4^j) = 2^j.$$

Proof. Place the base-4 digits of two integers on top of each other. Before carrying, the digit cost of their sum is exactly the sum of their two digit costs. A carry at position i replaces four units at that position by one unit at position $i + 1$. Its contribution to the digit cost therefore changes from $4 \cdot 2^i$ to 2^{i+1} and cannot increase. After all carries, the resulting cost is $c(r + s)$, proving

$$c(r + s) \leq c(r) + c(s).$$

If $4^J \leq r < 4^{J+1}$, then

$$c(r) \leq 3 \sum_{i=0}^{J} 2^i = 3(2^{J+1} - 1),$$

whereas $r \geq 4^J$. Thus $c(r)/r \leq 6/2^J$, which tends to zero. Finally, $4^j - 1$ has j base-4 digits, all equal to 3, while 4^j has a single digit 1 at position j. The two exact formulas follow. □

4 Full answer

Let $v_2(n)$ denote the exponent of 2 in the prime factorization of n, with $v_2(1) = 0$.

Theorem 2. *Define*

$$\phi(r) := r + c(r) \quad (r \in \mathbb{N}_0), \qquad \boxed{\omega(n) := 2^{\phi(v_2(n))} \quad (n \in \mathbb{N}).}$$

Then ω is a weight on the multiplicative semigroup $\mathbb{N}$, but it is not almost monotone. Furthermore,

$$\Omega(m, n) := \omega(m)$$

is a weight on $\mathbb{N}^2$ which is not almost monotone. Consequently item (4) of the source paper has an affirmative answer for both $\mathbb{N}$ and $\mathbb{N}^2$.

Proof. The function ϕ is nonnegative and subadditive by Lemma 1. Since $v_2(ab) = v_2(a) + v_2(b)$,

$$\omega(ab) = 2^{\phi(v_2(a)+v_2(b))} \leq 2^{\phi(v_2(a))+\phi(v_2(b))} = \omega(a)\omega(b).$$

Thus $\omega : \mathbb{N} \to [1, \infty)$ is a weight.

It is not admissible. Indeed, Lemma 1 gives

$$\lim_{r\to\infty} \omega(2^r)^{1/r} = \lim_{r\to\infty} 2^{(r+c(r))/r} = 2 > 1.$$

It remains to rule out the other branch of almost monotonicity. For $j \geq 1$, put

$$a_j = 2^{4^j-1}, \qquad b_j = 2^{4^j}.$$

Then $a_j \mid b_j$, while the exact formulas in Lemma 1 give

$$\begin{aligned}\phi(4^j-1) - \phi(4^j) &= (4^j-1) + 3(2^j-1) - (4^j+2^j) \\ &= 2^{j+1} - 4.\end{aligned}$$

Hence

$$\frac{\omega(a_j)}{\omega(b_j)} = 2^{2^{j+1}-4} \longrightarrow \infty.$$

No constant K can satisfy $\omega(a) \leq K\omega(b)$ whenever $a \mid b$. Because ω is also nonadmissible, it is not almost monotone.

For $\Omega(m,n) = \omega(m)$, submultiplicativity follows immediately from that of ω. It is nonadmissible because

$$\lim_{r\to\infty} \Omega(2^r, 1)^{1/r} = 2.$$

Finally, $(a_j, 1)$ divides $(b_j, 1)$ coordinatewise and

$$\frac{\Omega(a_j,1)}{\Omega(b_j,1)} = \frac{\omega(a_j)}{\omega(b_j)} \longrightarrow \infty.$$

Thus Ω fails the uniform almost-monotonicity inequality even on this one-coordinate chain. This proves the claim on $\mathbb{N}^2$. $\square$

5 Verification, scope, and novelty check

The proof has three independent audit points. First, base-4 carrying proves subadditivity of c. Second, $c(r) = o(r)$ makes the linear term determine the nonadmissible spectral growth. Third, the explicit pair of consecutive exponents $4^j - 1, 4^j$ gives an unbounded downward ratio along divisibility. The lift to $\mathbb{N}^2$ preserves all three properties.

A bounded exhaustive check verified $c(r+s) \leq c(r) + c(s)$ for all $0 \leq r, s \leq 1000$ and verified the exact jump formula for $1 \leq j \leq 12$. This is only a sanity check; the carry argument is the proof.

The page-15 multidimensional restatement mentions $K > 0$ but omits K from the following displayed comparison, whereas the formal $\mathbb{N}^2$ definition on page 2 includes it. The construction above has ratios tending to infinity, so it fails both the literal monotone comparison and the more permissive uniform-K comparison. The disjunction "admissible or uniformly monotone" is also handled explicitly: the weight fails both branches.

A bounded novelty check through 26 August 2026 searched the run's registry, solution, attempt, and proof-gap indexes; the exact phrase "not an almost monotone weight"; the source title, author, and arXiv id; and arXiv searches combining "almost monotone", "weight", and "Dirichlet". The search found the source preprint and its 2026 journal publication, but no independent paper answering item (4). The construction is elementary, so this search does not establish historical priority beyond those bounds. Novelty confidence is therefore moderate, while mathematical confidence is high.

The highest-value specialist check is to verify the carry argument and the source-definition match. Under either natural reading of the page-15 wording, the proof is complete.

References

[1] P. A. Dabhi, *On inversion of absolutely convergent weighted Dirichlet series in two variables*, arXiv:2407.19982 (2024); Mathematica Bohemica, first online 27 January 2026.